

TECHNICAL RESEARCH

Technical Research Documentation

Section 2.3 (b)

Voice Interaction for Spatial Context

How do you talk to a spatial AI? Real-time speech-to-speech with environmental context, scene understanding in voice pipelines, latency requirements, and existing approaches.

Version 1.0 | April 2026 | Internal Research Document

1. Introduction

Voice interaction in spatial computing differs fundamentally from conventional chatbot design. When a user speaks to an AI embedded in an augmented or virtual environment, the system must simultaneously interpret what is being said, understand the physical or virtual scene the user inhabits, and respond in a way that is contextually grounded, emotionally appropriate, and low-latency enough to feel natural.

This section documents the technical foundations of voice-to-voice interaction in spatial contexts, covering pipeline architectures, latency constraints, scene-context injection methods, and the landscape of production-grade models and APIs available as of 2026.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026

2. Why Voice Is the Natural Interface for Spatial AI

Speech is the primary mode of human-to-human communication. Several properties make it especially well-suited to spatial AI:

- **Speed:** An average person speaks at 125 to 175 words per minute, significantly faster than typed input.
- **Cognitive naturalness:** Speech is acquired before literacy, typically between 12 and 18 months of age, making it a lower-cognitive-load modality.
- **Accessibility:** Voice interfaces are inherently more inclusive for users with visual impairment, motor disabilities, or limited literacy.
- **Hands-free operation:** In mixed-reality contexts, users often cannot interact with a keyboard or touch surface, making voice the only practical input channel.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026

3. Pipeline Architectures for Voice Interaction

3.1 Generation 1: Text-Centric Chatbots

The earliest conversational AI systems use a simple three-stage pipeline: text input, LLM reasoning, text output. These systems offer high accuracy, easy logging, and noise robustness because they operate entirely in the text domain. However, they require literacy, have slow input throughput, and discard all emotional or prosodic context, making extended voice conversations feel unnatural.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026

3.2 Generation 1.5: Cascaded Voice Pipelines (ASR + LLM + TTS)

Generation 1.5 systems add Automatic Speech Recognition (ASR) at the input and Text-to-Speech (TTS) at the output, wrapping an LLM core. The pipeline follows a sequential flow: Voice → STT → LLM → TTS → Voice, where each component waits for the previous one to finish before processing. While more natural than pure text systems, they introduce three critical problems:

- **Latency accumulation:** Each component in the cascade adds processing time. Typical end-to-end latency in cascaded pipelines reaches approximately 2,500 milliseconds, far above the 200–300 ms threshold at which silence feels uncomfortable to human listeners.
- **Prosodic information loss:** Converting speech to text strips out tone, emotion, pacing, and other non-verbal signals. The LLM only processes the transcribed words, and the TTS layer regenerates a generic voice, losing the speaker's original emotional context entirely.

- **Error propagation:** Mistakes made by the ASR layer cascade downstream, degrading LLM response quality and ultimately TTS output.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Latency and Information Loss in Cascaded Pipelines

3.3 Generation 2: Native Speech-to-Speech (S2S) Models

Native S2S models bypass cascaded pipelines by directly processing and generating audio tokens end-to-end. This architecture offers three core advantages relevant to spatial AI:

- **Reduced latency:** Eliminating ASR and TTS components removes 600 to 800 milliseconds of processing time from the pipeline, enabling responses that approach human conversational rhythm.
- **Preserved prosody and emotional intelligence:** Because the model operates on audio tokens directly, it retains tonal and emotional information throughout processing, enabling it to match and respond to the emotional register of the speaker.
- **Full-duplex capability:** Native S2S models can listen, process, and generate audio simultaneously, allowing users to interrupt the model mid-response, mirroring the natural dynamics of human conversation.

To manage the high sample rate of audio (typically 16,000 samples per second), these models use audio tokenization and compression — specifically residual vector quantization (RVQ) — to reduce audio to a manageable rate of approximately 50 to 100 tokens per second.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Native S2S Model Advantages

3.4 Cascaded vs. Native S2S: Comparison

Feature	Cascaded Pipeline	Native S2S
Architecture	ASR → LLM → TTS (sequential)	Single model, audio in / audio out
Latency (typical)	800ms – 2,500ms end-to-end	200ms – 300ms Time to First Token
Prosody Preservation	Lost at ASR stage	Retained throughout
Full-Duplex Support	Difficult; requires barge-in logic	Native capability
Modularity	High; components swappable	Low; single model
Debuggability	High; intermediate text inspectable	Low; no intermediate representations
Cost (long sessions)	Lower	Higher (context window accumulates)

Source: [Softcery, January 2026](#)

Source: [Introl, January 2026](#)

Source: Internal Research Meeting, February 2026

3.5 Half-Duplex vs. Full-Duplex Communication

Mode	Description	Suitability for Spatial AI
Half-Duplex	Model and user cannot speak simultaneously. Often requires push-to-talk.	Poor. Spatial interactions require fluid, interruptible conversation.
Full-Duplex	Model listens, processes, and generates output simultaneously. Users can interrupt freely.	Preferred. Enables natural, real-time conversation with a spatial AI character.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Types of Communication in S2S Systems

4. Latency Requirements for Spatial Voice Interaction

Latency is the single most critical quality metric for voice interaction in spatial contexts. The following thresholds define the boundaries of acceptable performance.

4.1 Human Conversational Threshold

According to human conversational benchmarks, the ideal turn-taking delay is approximately 200 milliseconds. Delays beyond this threshold disrupt the natural flow, causing users to feel as though the system is unresponsive or inattentive.

When someone speaks to an AI system, they expect a response within 300 milliseconds — the natural pause length in human conversation. Delays over 500ms make users want to repeat themselves or give up entirely.

Source: [VideoSDK, 2025](#)

Source: [Deepgram, 2026](#)

4.2 Industry Latency Benchmarks

Production voice AI agents typically aim for 800ms or lower end-to-end latency to maintain conversational flow. Current models achieve Time to First Token (TTFT) in the 200–300 millisecond range: Google's Gemini Flash logs approximately 280ms, and OpenAI's GPT-4o Realtime achieves approximately 250–300ms.

Even pauses as short as approximately 300 milliseconds can feel unnatural, while any latency beyond approximately 1.5 seconds can rapidly degrade the experience. Achieving sub-second responsiveness requires deep optimizations across the entire system, from networking to speech recognition, large language models, and TTS.

Source: [Retell AI](#)

Source: [Cresta, October 2025](#)

4.3 Component-Level Latency Breakdown (Cascaded Architecture)

In a cascaded pipeline, latency accumulates as follows: audio travels from the user device to the media edge (40ms) with buffering (30ms) and decoding (25ms), then through orchestration to STT (350ms), LLM (375ms), and TTS (100ms), each separated by brief service hops of approximately 10ms.

- **Speech-to-Text:** 100–500ms depending on streaming configuration.
- **LLM:** 350ms for optimized models to over 1,000ms for frontier models.
- **Streaming TTS:** 75–200ms time-to-first-audio.

Source: [Introl, January 2026](#)

4.4 Latency Reference Table

Latency Threshold	User Perception	Classification
< 200ms	Imperceptible; feels instant	Target (human-equivalent)
200 – 300ms	Acceptable; natural conversation	Minimum production requirement
300 – 800ms	Noticeable; tolerable in enterprise settings	Operational floor
800ms – 1,500ms	Disruptive; users lose confidence	Risk threshold
> 1,500ms	Unacceptable; conversation breaks down	Failure condition
~2,500ms	Clearly robotic and frustrating	Cascaded pipeline average (Internal Research, Feb 2026)
800ms – 1,200ms	Breaks natural rhythm in continuous dialogue	ElevenLabs with 3+ tool calls
< 600ms	Required for immersive spatial AI	Native S2S target (Internal Research)

Key Insight: The 200–300ms uncomfortable silence threshold is well-established in human communication research. Any voice system for spatial AI must target sub-600ms end-to-end latency to maintain immersion.

Voice Activity Detection (VAD) has been completed and Voice Activity Detection (VAD) has been completed and a native S2S model architecture was selected specifically to reduce latency and eliminate multi-layer server overhead. On-device model alternatives are currently under active research and development to reduce delay further.

Source: *Internal Monthly Progress Report, November 2025 — Live Conversations and Voice I/O*

4.5 Gemini Flash-Lite: Latency Profile for Asynchronous Use

For audio emotion detection workflows using Gemini Flash-Lite models, inference time after server-side file preparation stabilizes at approximately 5 seconds regardless of audio length. This makes it unsuitable for real-time voice loops but viable for asynchronous analysis pipelines.

Audio Length	Gemini 2.5 Flash-Lite Processing Time	Gemini 2.0 Flash-Lite Processing Time
15 – 20 seconds	4.7 seconds	4.1 seconds
29 – 41 seconds	5.0 – 5.2 seconds	5.0 seconds

Audio Length	Gemini 2.5 Flash-Lite Processing Time	Gemini 2.0 Flash-Lite Processing Time
70 – 75 seconds	5.0 seconds	4.9 seconds

Note: Server-side file pre-processing adds an unpredictable additional delay. A 1-minute WAV file can take approximately 2 minutes for server-side preparation, making these models unsuitable for real-time voice interaction in spatial contexts.

Source: *Internal Audio Processing and Emotion Analysis Research — Processing Results and Bottleneck Analysis*

5. Scene Understanding and Environmental Context in Voice Pipelines

5.1 The Core Challenge

A spatial AI must not only understand what a user says but also the environment in which the conversation is taking place. Scene context encompasses:

- Physical or virtual surroundings — objects, surfaces, and spatial layout
- User location and gaze direction
- Nearby entities, characters, or points of interest
- Ambient acoustic conditions — noise level, reverb, and speaker overlap
- Session history and current task state

The inability to reason about structured 3D environments constrains utility in spatially grounded tasks such as embodied AI, immersive simulations, and interactive 3D applications.

5.2 Context Injection Methods

Scene understanding data must be injected into the voice pipeline at the appropriate layer. The method depends on whether the system uses a cascaded or native S2S architecture.

5.2.1 System Prompt Architecture (Cascaded and Hybrid Systems)

In cascaded pipelines, scene context is injected as structured text in the LLM system prompt before each turn. This includes a serialized representation of the environment state, current objects in view, user location, and task context. The model processes this alongside the transcribed user utterance.

ElevenLabs Agents use this approach. The agent receives a configurable system prompt that can embed environmental metadata. However, adding more than two to three heavy tool calls or complex reasoning tasks in this prompt causes the LLM to enter sequential processing mode, pushing end-to-end latency above 800 milliseconds to 1.2 seconds, which is noticeable in real-time conversation.

Source: *Internal R&D: 11Labs Agents Real-Time Speech-to-Speech — Maximum Custom Functions Without Increasing Latency*

5.2.2 Hybrid System Prompt Architecture (Native S2S Models)

The Persona model, an open-source native S2S model built on top of Moshi, implements a hybrid system prompt architecture that allows both text-based context and audio samples to be passed into the model simultaneously. This enables:

- **Text-side:** Injection of scene state, character persona, and spatial metadata.
- **Audio-side:** Voice cloning reference samples to maintain character voice consistency across sessions.

This dual-channel system prompt design is particularly relevant for spatial AI, where the character voice must remain consistent while scene context changes dynamically.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Moshi and Persona Model Architectures

5.2.3 Structured Context via Scene Graph and Point Cloud

A new paradigm enables vision-language models to generate, understand, and edit complex 3D environments by injecting a continually evolving spatial context. Constructed from multimodal input, this context consists of three components:

- **Scene portrait:** A high-level semantic blueprint of the environment.
- **Semantically labeled point cloud:** Captures object-level geometry.
- **Scene hypergraph:** Encodes rich spatial relationships, including unary, binary, and higher-order constraints.

Together, these components provide the model with a structured, geometry-aware working memory that integrates inherent multimodal reasoning capabilities with structured 3D understanding for effective spatial reasoning.

5.2.4 Retrieval-Augmented Generation (RAG) for Contextual Grounding

WavRAG is a pioneering audio-integrated RAG framework that enables spoken dialogue models to retrieve and utilize both audio and text knowledge bases, with end-to-end audio support for real-time, context-aware conversation. This approach is applicable to spatial AI systems where historical scene knowledge must be retrieved dynamically.

5.2.5 Function Calls and Tool Integration

The OpenAI Realtime API supports remote MCP servers, image inputs, and SIP phone calling, making voice agents more capable through access to additional tools and context. Function calling makes it possible for voice assistants to respond to user requests by triggering actions or pulling in new context — for example, retrieving relevant environmental information to personalize responses.

Source: [OpenAI Realtime API Documentation](#)

5.3 Asynchronous Context Pipeline

For heavier scene-understanding tasks — such as AR plane detection, surface mapping, object recognition, and user gaze tracking — the recommended pattern is to offload these computations to an asynchronous backend. The results are then injected into the voice pipeline as a structured state object before the next conversation turn begins, keeping the real-time voice loop clean and low-latency.

This separation of concerns allows:

- The real-time voice channel to maintain low latency without blocking on complex spatial computation.

- The scene understanding backend to operate at its own cadence, pushing updates into the context store.
- Graceful degradation when spatial data is unavailable or stale.

Source: *Internal R&D: 11Labs Agents Real-Time Speech-to-Speech — Practical Recommendation*

5.4 Acoustic Context and Noise Handling

Spatial environments introduce acoustic challenges not present in standard voice interfaces. Real-world noise, reverb from surfaces, overlapping speakers, and microphone positioning all degrade audio quality and model confidence. These factors are identified as active areas of evaluation for native S2S models in production:

- Real-world noise: Background sound in physical or mixed-reality environments.
- Conversational drift: Topic shifts over long sessions that degrade response coherence.
- Speaker overlap: Multiple users speaking simultaneously.
- Safety boundary enforcement in spatially ambiguous scenarios.

Unity's default microphone interface lacks noise cancellation and advanced audio processing. In the current implementation, a Native Android SDK has been integrated alongside Unity to address this gap, providing hardware-level audio preprocessing before audio reaches the S2S model.

Source: *Internal Monthly Progress Report, November 2025 — Native Audio Stack Upgrade*

5.5 Spatial Context in the Current Implementation

In the this project (November 2025 Monthly Progress Report), accuracy improvements in AR plane and surface detection were noted as ongoing work. Surface detection results feed directly into the spatial state of the AI character and must be available to the voice system in near-real-time so that spoken responses are grounded in the current physical environment of the user.

Source: *Internal Monthly Progress Report, November 2025*

6. Existing Approaches and Models

6.1 Proprietary Real-Time S2S APIs

6.1.1 OpenAI Realtime API

OpenAI's Realtime API provides a low-latency, multimodal speech-to-speech endpoint. The API uses a persistent WebSocket connection. It supports MCP servers, image inputs, and SIP phone calling. Two model tiers are available:

Model	Input (per min)	Cached Input (per min)	Output (per min)	Total (per min)
gpt-realtime	\$0.026	\$0.00033	\$0.053	~\$0.080
gpt-realtime-mini	\$0.0083	\$0.00025	\$0.0167	~\$0.025

Cost scenarios:

Scenario	Duration	gpt-realtime Cost	gpt-realtime-mini Cost
Voice chat (2 min each side)	4 min	~\$0.32	~\$0.10
Half-hour conversation	30 min	~\$2.40	~\$0.76
One-hour conversation	60 min	~\$4.80	~\$1.52

Important cost note: Tokens accumulate in the context window, and the model re-charges for all previous tokens on each turn, making longer conversations disproportionately more expensive.

The gpt-realtime model demonstrates higher intelligence and can capture non-verbal cues such as laughter, switch languages mid-sentence, and adapt tone. It is better at interpreting complex system messages and developer prompts, including reading scripts word-for-word, repeating alphanumerics, and switching languages seamlessly.

Source: [OpenAI Realtime API Pricing Documentation](#)

6.1.2 ElevenLabs Agents

ElevenLabs Agents provide a modular pipeline that processes audio input through LLM logic to speech output, with optional custom tool attachment. Three voice model tiers are available for real-time use:

Model	Characteristic	Best Use Case
Multilingual v2	High quality, expressive	Character-driven spatial interactions
Turbo v2.5	Balanced quality and speed	Continuous real-time conversations
Flash v2.5	Fastest available	Ultra-low-latency turn-taking

Subscription plans:

Plan	Price/Month	Agent Minutes	Suitable For
Free	\$0	~15 min	Testing / proof of concept
Starter	\$5	~50 min	Small projects
Creator	\$22	~250 min	Pro cloning, higher audio quality
Pro	\$99	~1,100 min	High throughput API use
Scale	\$330	~3,600 min	Teams and startups at scale
Enterprise	Custom	Custom	Dedicated infrastructure, real-time guarantees

R&D finding: Adding more than three tool calls or heavy custom functions to an ElevenLabs agent causes latency to jump to 800ms–1,200ms, breaking conversational rhythm. The recommendation is to keep the agent focused on conversation and offload heavier logic (analytics, database operations) to an asynchronous external backend.

Source: Internal R&D: 11Labs Agents Real-Time Speech-to-Speech

6.1.3 Other Proprietary Options

Additional proprietary native S2S models include:

- **Google Gemini Live API:** Google's Gemini Flash logs approximately 280ms TTFT in controlled settings.
- **Amazon Nova:** Supports S2S interaction with varying levels of language coverage.
- **Hume AI:** Focuses on emotional intelligence in voice interaction.

These options involve cloud dependency and per-usage costs that must be evaluated against the on-device strategy.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Proprietary and Open-Source Native S2S Models

6.2 Open-Source Native S2S Models

6.2.1 Moshi

Moshi is a research-focused, open-source native S2S model designed for full-duplex, real-time conversations. Its architecture includes:

- An audio encoder that converts user speech into discrete audio tokens.
- A temporal transformer that generates semantic and acoustic output tokens.
- An optional inner monologue mechanism for internal reasoning.

Moshi demonstrates a real-time factor faster than real-time processing, meaning it can generate responses faster than the duration of the audio it produces. GPU requirements are approximately 18 GB of VRAM. Quantized model variants show better performance and more processing headroom.

6.2.2 Persona

Persona is built on the Moshi architecture with key enhancements for production deployment:

- Hybrid system prompt architecture supporting both text context and audio reference samples.
- Voice cloning capability via audio sample injection.
- More coherent response generation and fewer speech segments per query compared to Moshi.

GPU utilization for Persona runs at approximately 70%, consuming around 18 GB of VRAM. The model is suitable for self-hosted deployment with fixed infrastructure costs.

Metric	Moshi	Persona
Architecture	Audio encoder + temporal transformer	Moshi base + hybrid system prompt
Full-Duplex	Yes	Yes
Voice Cloning	No	Yes
Real-Time Factor	Faster than real-time (quantized)	Faster than real-time (quantized)
VRAM Required	~18 GB	~18 GB

Metric	Moshi	Persona
Self-Hosting Complexity	High	High

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Moshi and Persona Architectures and Performance Metrics

6.3 Cascaded Architecture Pipelines (Reference Approach)

The sequential pipeline (VAD → STT → LLM → TTS) remains the standard architecture for production voice agents where modularity and debuggability are priorities. Streaming at every stage boundary is what makes the difference between a 2-second response and a 400ms response. The cascaded pipeline gives modularity, debuggability, and provider flexibility that S2S models cannot yet match.

A low-latency, end-to-end voice-to-voice communication pipeline integrates streaming ASR, a quantized LLM, retrieval-augmented generation, and real-time TTS to enable responsive, knowledge-grounded spoken interactions.

7. Audio Emotion Detection for Contextual Response

Emotion detection from raw audio allows the spatial AI to understand the user's affective state and adjust its response accordingly, improving conversational naturalness and emotional intelligence.

7.1 OpenAI Audio Models for Emotion Detection

Two OpenAI audio-capable models were evaluated for detecting emotions with timestamps from raw audio:

Model	Latency	Strengths	Limitations
gpt-4o-audio-preview	8 – 12 seconds	Strong audio comprehension. Higher accuracy when successful.	Not emotion-specialized. Sometimes refuses emotion tasks. Inconsistent JSON format.
gpt-audio-mini-2025-10-06	3 – 6 seconds	Very fast. Cost-effective.	Low accuracy. Poor timestamp detail. Frequent JSON format violations.

7.2 Consistency Challenges

Neither model is consistent enough for production-grade emotion detection without additional post-processing. Key causes of inconsistency include:

- **General-purpose architecture:** These models are designed for broad audio understanding, not psychological emotion inference specifically.
- **Audio quality sensitivity:** Noise, flat speaking tone, low volume, or monotone delivery reduces model confidence and triggers refusals.

- **JSON enforcement limitations:** Even with strict system-prompt instructions specifying JSON-only output, models occasionally respond with natural language explanations, partial JSON, or invalid structures.

A dedicated emotion-analysis model or a strong post-processing validation layer is required for stable real-world use.

Source: *Internal Audio Processing and Emotion Analysis Research — Conclusion and Consistency Challenges*

8. Voice Cloning for Spatial AI Characters

Spatial AI characters require a consistent, distinctive voice that persists across sessions and can be adapted to regional accents or language variants. Voice cloning allows a defined character voice to be reproduced and applied to dynamically generated speech output.

Four models were evaluated in the internal Voice Cloning R&D study using five English voice samples (34 seconds or fewer each, WAV format). Providing reference text alongside reference audio was identified as the most significant factor for cloning quality across all models.

8.1 Model Evaluation Summary

Model	Platform	Assessment	Key Notes
Sesame CSM 1B	Google Colab (T4 GPU)	Recommended	Highest accuracy and voice similarity. Natural tone. Fastest generation on Colab.
Qwen3TTS (0.6B / 1.7B)	Local / HuggingFace	Second-best	Strong results with reference text. 2–4 min per voice locally. Runs on CPU.
CosyVoice 3	Local	Not acceptable	Errors during output generation. Distorted audio. Testing discontinued.
Fish Speech	Google Colab	Not acceptable	Distorted and glitchy output. High processing time.

8.2 Runtime Comparison: Sesame CSM 1B vs. Qwen3TTS

Voice	Qwen3TTS Clone Time (min)	Qwen3TTS Total (min)	Sesame Clone Time (min)	Sesame Total (min)
Chris Tucker	3.82	4.00	0.95	1.73
Heath Ledger	2.17	2.34	0.75	1.52
Imran Khan	4.03	4.18	1.70	4.08
Barack Obama	3.90	4.06	0.93	1.70
Peter Dinklage	3.38	3.60	1.08	1.84

Note: Sesame CSM 1B was tested on Google Colab with a T4 GPU. Total time includes model loading on first run. Sesame CSM 1B was significantly faster across all voices.

Sesame CSM 1B is available for testing at: <https://huggingface.co/spaces/sesame/csm-1b>

Qwen3TTS is available at: <https://huggingface.co/spaces/Qwen/Qwen3-TTS>

Source: *Internal Voice Cloning R&D Report — Runtime Comparison and Model Evaluation*

8.3 Spatial AI Use Cases for Voice Cloning

- **Regional accent personalization:** Adapting a character's voice to the region where the application is deployed.
- **Brand voice consistency:** Cloning a defined AI character voice for use across all interaction sessions.
- **Multilingual continuity:** A creator records in one language; the cloned voice speaks translated versions while preserving original cadence and timbre.
- **Narrative memory characters:** Cloning voice samples from recordings to enable users to interact with a familiar voice through a conversational avatar.

Source: *Internal Voice Cloning R&D Report — Use Cases*

9. Lip Sync and Emotion Animation Integration

In spatial AI, voice output must be synchronized with character animation. Two subsystems are relevant: lip sync (viseme mapping) and facial emotion expression.

9.1 Lip Sync: Salsa

Salsa has been selected as the default on-device lip-sync model in the current implementation. Key advantages over evaluated alternatives (OVR and Rubab) include near-zero latency, no server cost, and superior performance. A commercial Salsa license is required to unlock full 7-blend-shape support; the current test implementation is limited to 4 blend shapes.

Source: *Internal Monthly Progress Report, November 2025 — Lip Sync and Facial Expressions*

9.2 Emotion Engine

A local dictionary-based emotion engine has replaced AI-driven real-time emotion switching, which caused jarring and unpredictable animation transitions. The current implementation uses:

- Context-based emotional state detection from conversation content.
- Timed emotion triggers firing approximately every 4 to 5 seconds.
- Smooth blend transitions between emotional states.

Source: *Internal Monthly Progress Report, November 2025 — Emotion Engine Refactor*

10. Limitations and Open Problems

The following areas require further research and real-world validation before native S2S voice interaction in spatial contexts can be considered production-ready:

Problem Area	Description
Real-world noise	Spatial environments introduce unpredictable acoustic noise that degrades ASR and S2S model performance.
Conversational drift	Long sessions cause topic shift and context loss, reducing response coherence over time.
Speaker overlap	Multiple users speaking simultaneously is not reliably handled by current open-source S2S models.
Safety boundary enforcement	Spatially ambiguous or adversarial inputs may cause native S2S models to violate safety constraints in ways not observed in text-only models.
Emotion detection reliability	No current general-purpose audio model provides consistent, timestamped emotion output suitable for production use.
Self-hosting compute cost	Moshi and Persona each require approximately 18 GB of VRAM. Scaling to multiple concurrent users requires significant GPU infrastructure.

Source: Internal Research Meeting: Speech-to-Speech Modeling, February 2026 — Model Evaluation Requires Real-World Testing

11. Architecture Recommendations for Spatial Voice Interaction

Based on the combined internal and external research reviewed, the following architecture guidance applies to voice interaction in spatial AI systems.

11.1 For Minimum Latency and Natural Conversation

Use a native S2S model (gpt-realtime, Gemini Live, or on-device options such as Persona) with spatial context injected through the system prompt and updated via function calls when scene state changes. Keep the agent pipeline lean; move all non-conversational logic to an asynchronous backend service.

11.2 For Maximum Control and Debuggability

Use a cascaded pipeline (VAD → STT → LLM → TTS) with scene data injected as structured text into the LLM system prompt. Where accuracy and domain control are paramount, a cascaded pipeline allows use of specialized ASR tuned to the domain and permits careful post-processing of outputs.

11.3 For Turn-Taking and Interruption Handling

Use a semantic turn detection model that decides when the user has finished speaking based on context rather than silence alone. In common cases, this approach can reduce response times to under 300ms without cutting users off mid-utterance.

11.4 For Transport Protocol

WebRTC is the standard for low-latency audio and video streaming in browsers and mobile apps. It uses UDP for fast transmission and handles packet loss gracefully. Both OpenAI and Google use WebRTC for client-side audio capture and playback.

11.5 For Full-Duplex Behavior

Speech-to-speech architectures stream input and output concurrently across the stack, reducing perceived delay in scenarios that involve rapid turn-taking or mid-utterance interactions.

12. Summary and Recommendations

Decision Area	Recommended Approach
Voice pipeline architecture	Native Speech-to-Speech model. Eliminates ASR and TTS latency layers. Required for immersive spatial interaction.
Duplex mode	Full-duplex. Allows interruptions and natural conversational rhythm.
Scene context injection	Asynchronous backend pipeline for heavy spatial computations. Inject results as structured state into system prompt before each turn.
Latency target	Sub-600ms end-to-end. On-device models preferred. Cloud APIs (OpenAI Realtime, ElevenLabs Flash) acceptable for cloud-hosted deployments.
Voice cloning	Sesame CSM 1B (primary). Qwen3TTS 0.6B (fallback for local/offline). Always provide reference text with reference audio.
Emotion detection	Not production-ready with general-purpose models. Use asynchronous analysis for post-processing. Dedicated emotion model required for real-time use.
Lip sync	Salsa (on-device). Near-zero latency. Full commercial license required for 7-blend-shape support.
Microphone stack	Native Android SDK integrated alongside Unity. Provides hardware-level noise cancellation absent from Unity's default mic.

12.1 Summary of Key Findings

Topic	Finding	Source
Comfortable latency threshold	200ms turn-taking delay	VideoSDK, 2025
Production latency standard	800ms or lower	Retell AI
Cascaded pipeline total latency	~1,000ms typical; up to ~2,500ms	Introl, 2026 + Internal Research, Feb 2026
Native S2S TTFT	200–300ms	Softcery, 2026
Cascaded pipeline prosody loss	Confirmed	Deepgram
S2S model selected (Internal Research)	Speech-to-Speech (on-device target)	Internal Progress Report, Nov 2025

Topic	Finding	Source
Recommended voice cloning model	Sesame CSM 1B	Internal Voice Cloning R&D Report
Scene context injection method	System prompt + function calls	OpenAI Realtime API
Real-time emotion detection status	Not production-ready	Internal Audio Processing Research
Gemini Flash-Lite for real-time	Not suitable (5s+ inference + pre-processing delay)	Internal Audio Processing Research

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